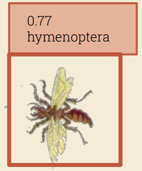
Building Computer Vision Pipelines for Eco-Evo Research

UIUC Machine Learning Workshop

Elizabeth Postema
August 2026



Workshop Structure

Part I:
Computer Vision Basics

Part II:
Using Roboflow to Make Models

Part III:
Building Pipelines (Code Demo)

Housekeeping

1. Log into your roboflow account
    a. Under settings > API keys > generate API Key

2. In the UIUC ML Github:
    a. Go to the computer_vison README
    b. Click the .zip link to download image sets

3. Also in the UIUC ML Github:
    a. Click the badge for the Colab notebook in the README

Translating (lots of) IMAGES → (lots of) ORGANIZED DATA

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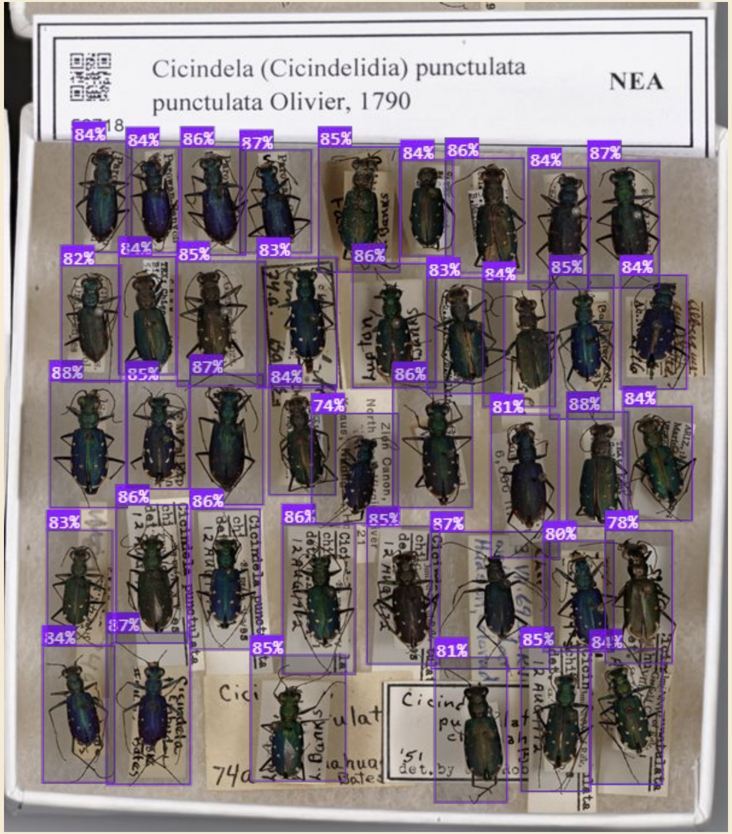


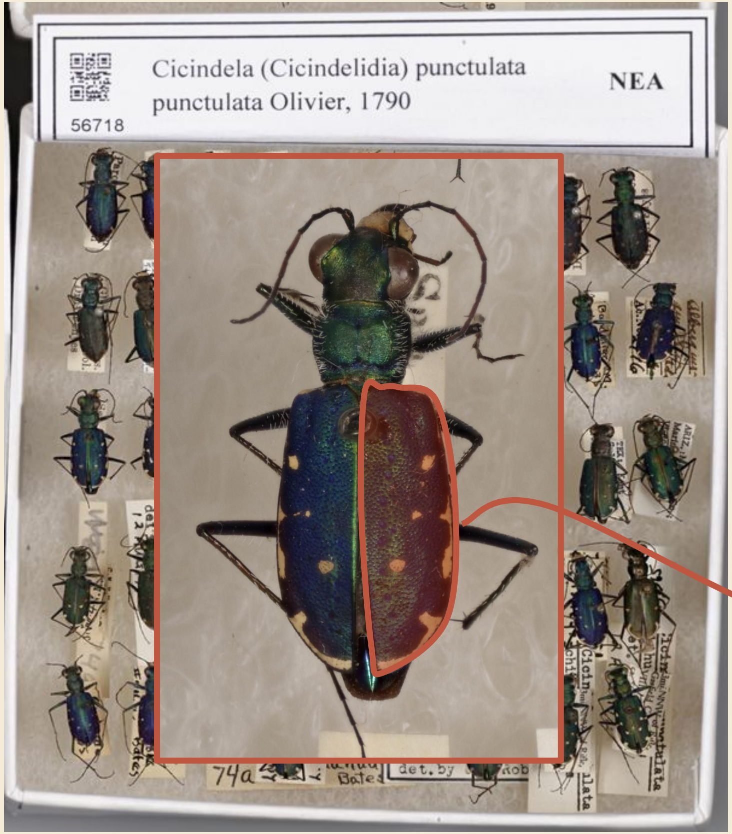
- Count stuff

Translating (lots of) IMAGES → (lots of) ORGANIZED DATA

- Count stuff
- **Identify some characteristics**

C. p. punctulata
female
damaged
etc...

Translating (lots of) IMAGES → (lots of) ORGANIZED DATA



- Count stuff
- Identify some characteristics
- **Measure stuff**

elytra width
% white vs. green

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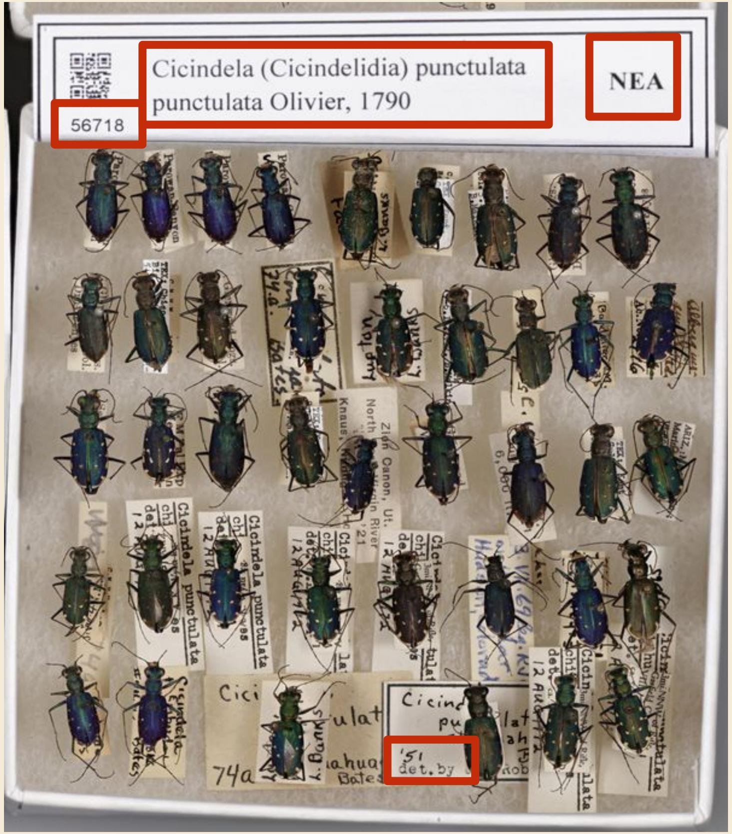

- Count stuff
- Identify some characteristics
- Measure stuff
- Transcribe text

Translating (lots of) IMAGES → (lots of) ORGANIZED DATA

- Count stuff
- Identify some characteristics
- Measure stuff
- Transcribe text

All (relatively) simple by hand…
but time-consuming at large scales

Translating (lots of) IMAGES → (lots of) ORGANIZED DATA

“I want to potentially **taxonomically sort** microscope photographs of insect specimens I collect in my field work”

“we have been collecting images with the "moth-o-matic," … which we hope to use for **abundance/biomass estimation** as well as **identification of live nocturnal insects**”

“ I have plant specimen photos of wetland and grassland vegetation, and would love to create models to do the following: 1. **Identify** dead vs. living plant matter; 2. **distinguish between different species of plants** to get some measure of plant diversity in a photo … 3. if possible, **identify plant species to the species or genus levels**.”

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Each one of these is a computer vision task

(or combo of CV tasks)

Main 3 Types of Computer Vision Models

CLASSIFICATION | DETECTION | SEGMENTATION

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American Goldfinch

Main 3 Types of Computer Vision Models

CLASSIFICATION

American Goldfinch

DETECTION

SEGMENTATION

Main 3 Types of Computer Vision Models

CLASSIFICATION

American Goldfinch

DETECTION

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Main 3 Types of Computer Vision Models

Multilabel Classification

1: American Goldfinch
2: Male 3: Breeding Plumage

DETECTION

SEGMENTATION

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Multilabel Classification

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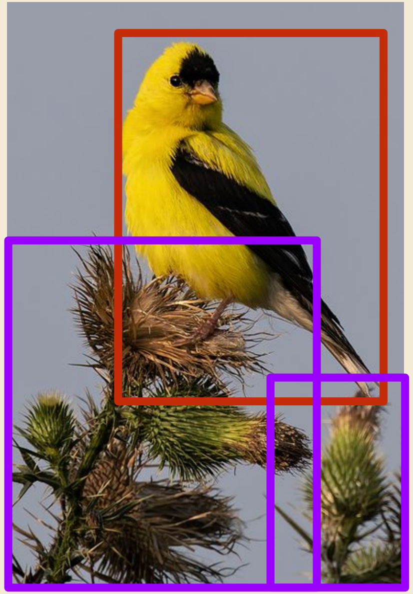
Detect multiple classes



1 Goldfinch
2 Thistles

SEGMENTATION

Main 3 Types of Computer Vision Models

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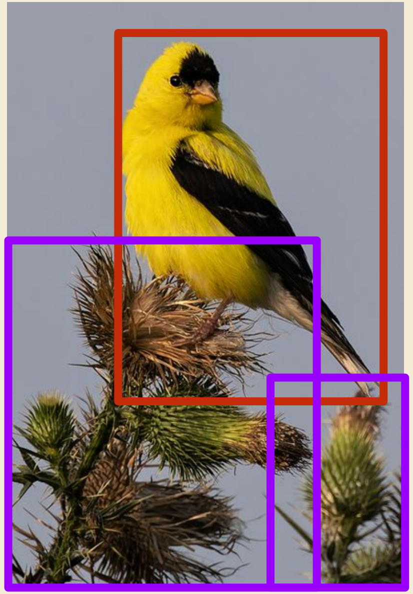

1 Goldfinch
2 Thistles

Segment Many Features

Head Body
Wing Tail

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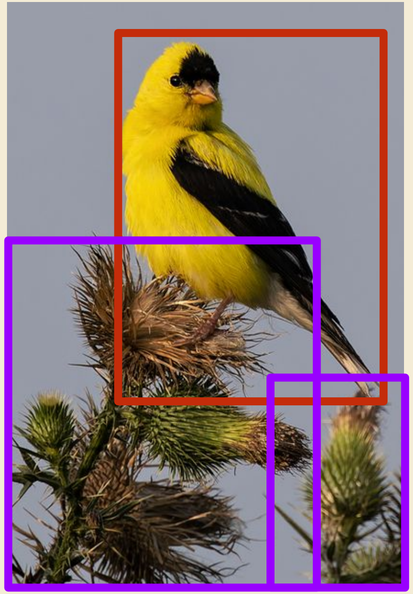

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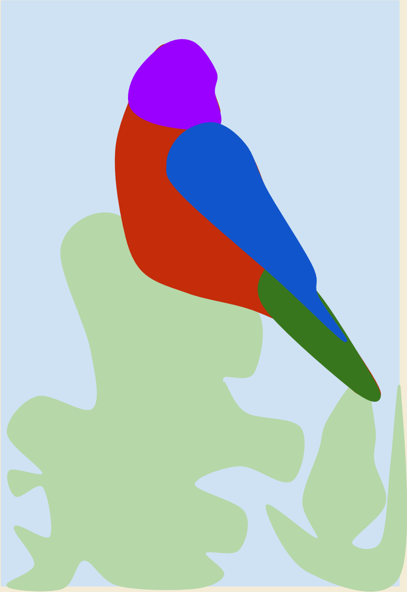
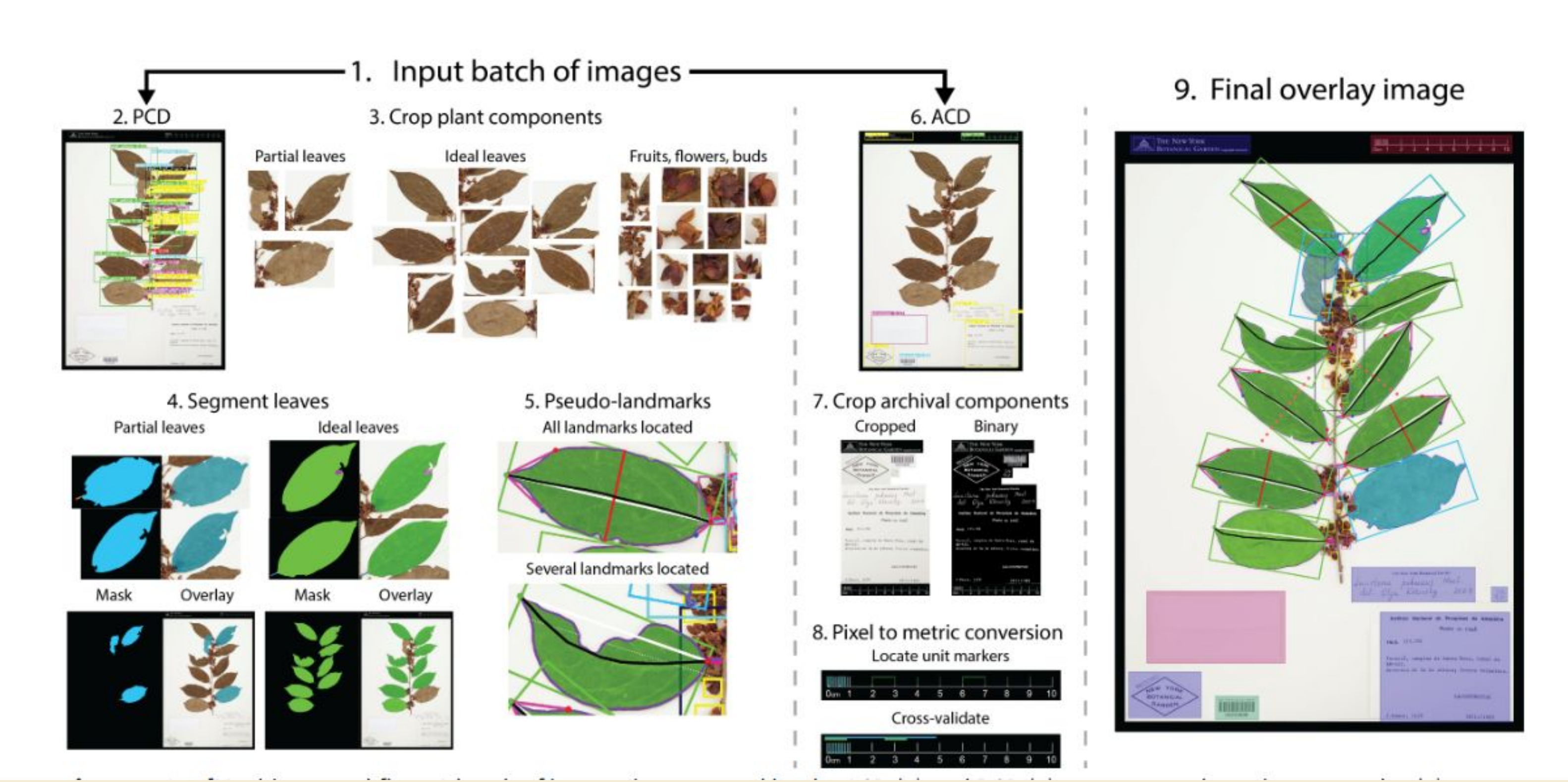
Semantic Segmentation



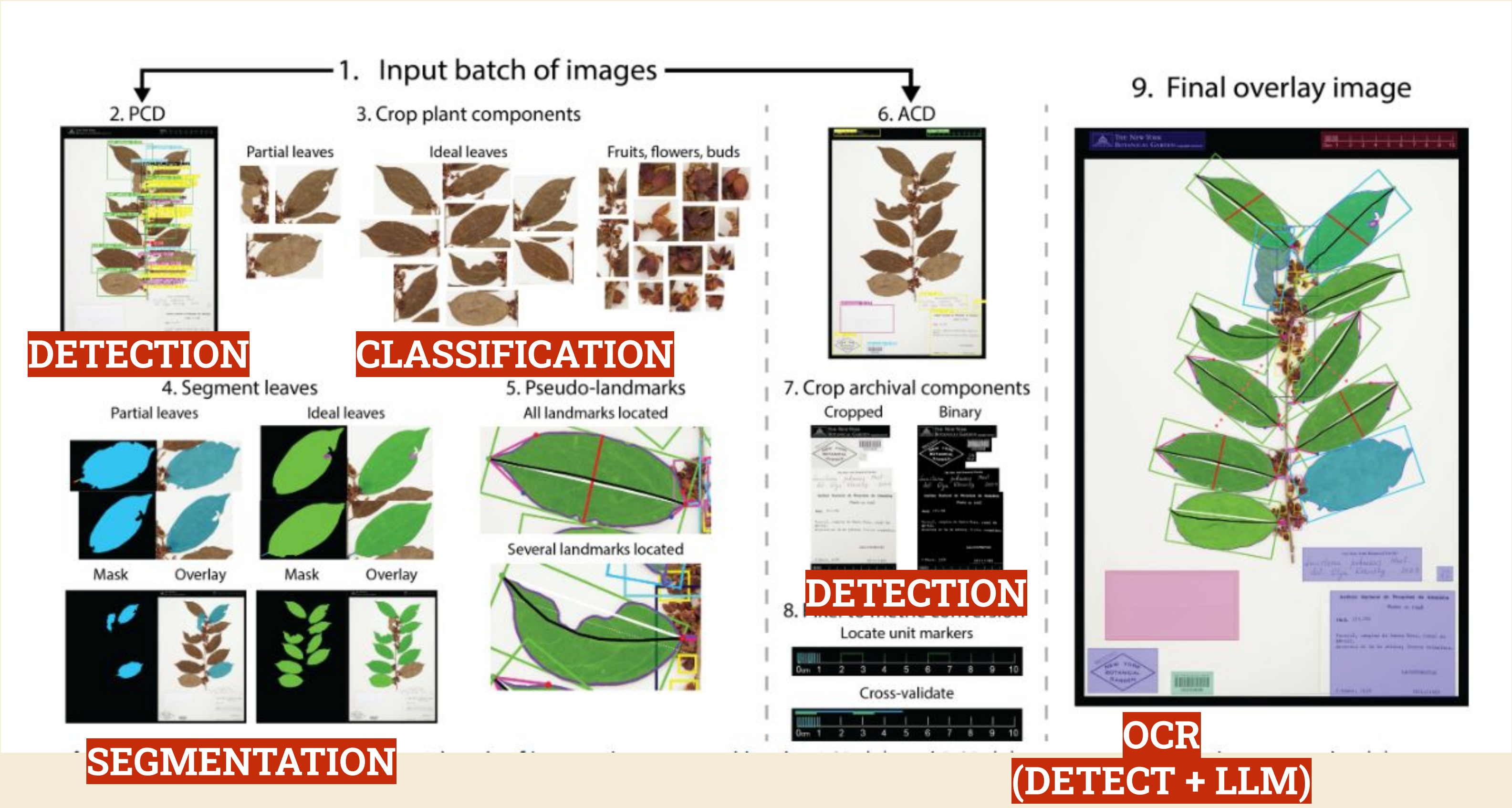
Label ALL pixels with a class

The Power of Combining Models: LeafMachine



Weaver et al. 2026 | leafmachine.org

The Power of Combining Models: LeafMachine



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Efficient Computer Vision Pipeline

PIPELINE-BUILDING GROUP ACTIVITY

1. Break into groups of 4-5

2. Think of a **LARGE IMAGE SET** you want to get **DATA FROM** to answer some eco-evo research question

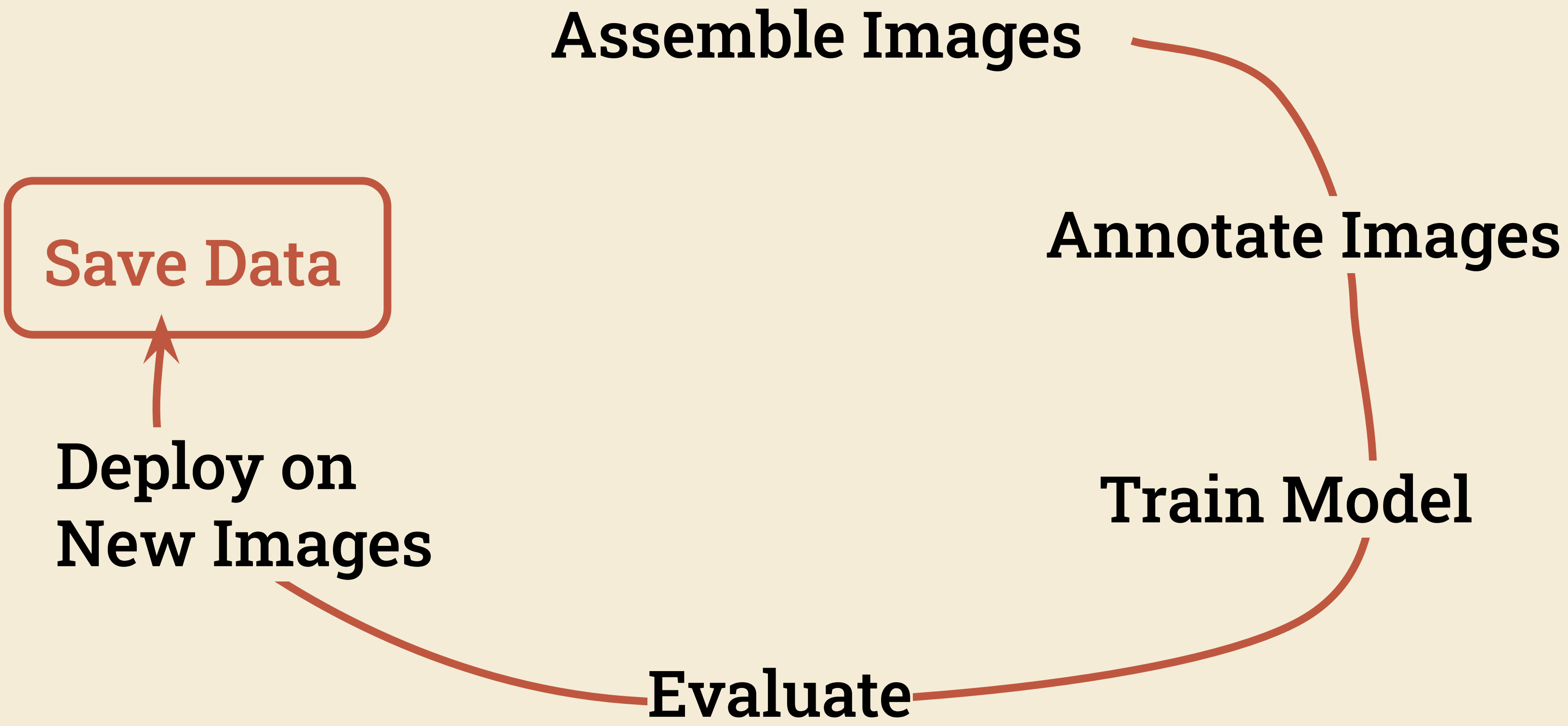
3. **DRAW OUT** the process to get from **IMAGE SET** → **DATA**
    a. Try to break down into AS MANY STEPS as you can
    b. Think in terms of specific **inputs**, **outputs**, and **processes**

4. Indicate **COMPUTER VISION**-based steps and model type

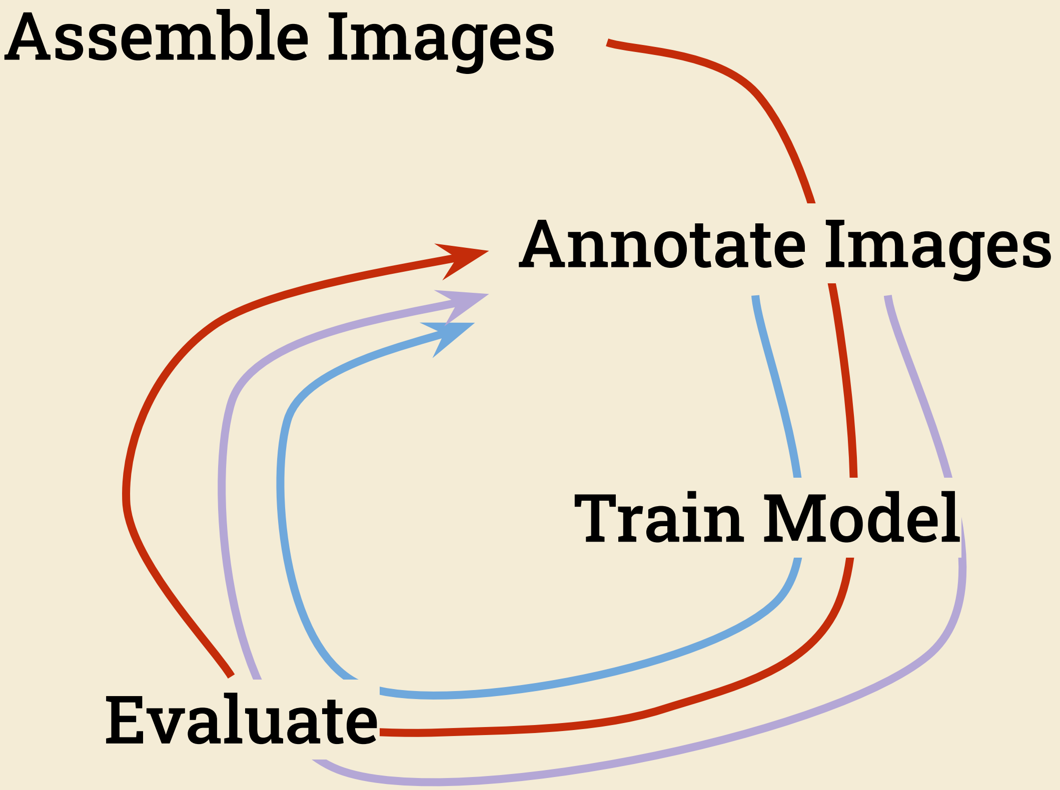
5. After **15 minutes,** a volunteer will share their group's design for discussion

TIP: There may be more than one way to build the pipeline!

CV Models: The Basic Loop



CV Models: The Basic *LOOPS*



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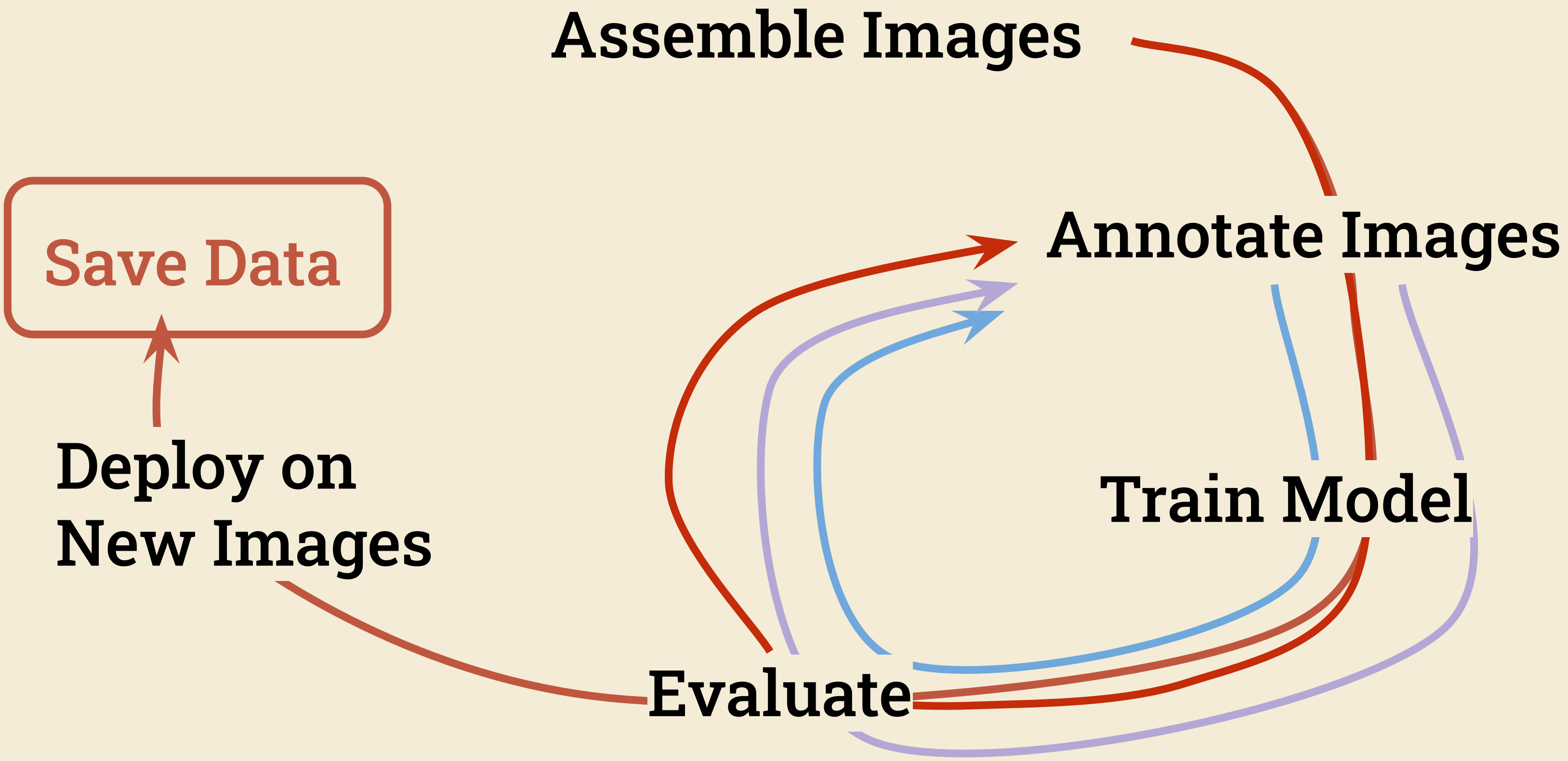

Assembling Images for Model Training

Images that TRAIN THE MODEL should MIRROR images from your FULL DATASET

Flowering

Fruiting

<u>Example</u>

I want a model that can identify *life stage* of swamp milkweed from iNat photos

Assembling Images for Model Training

Fruiting

Non-Flowering

Flowering

Budding

Flowering

Evenly represent
<u>all</u> classes
(including edge cases)

Assembling Images for Model Training


Fruiting


Non-Flowering


Flowering


Budding
Flowering

Evenly represent all classes
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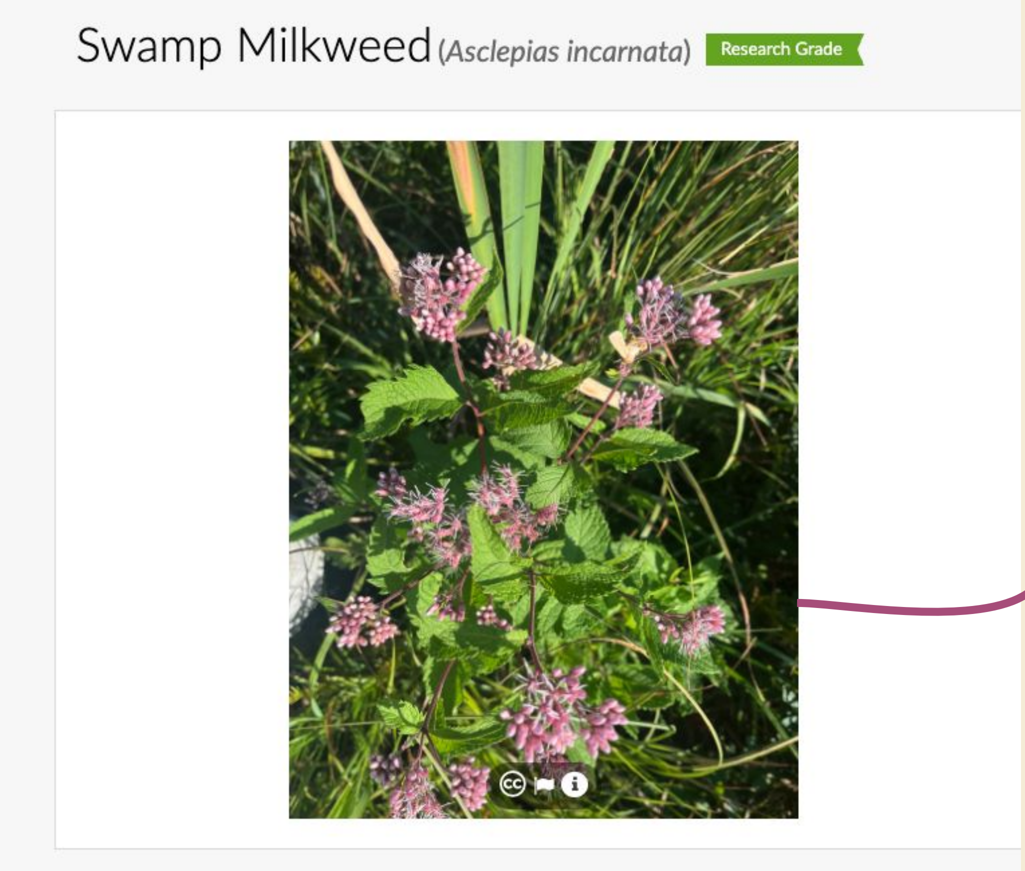
Flowering


Flowering


Flowering

Represent realistic within-class variation

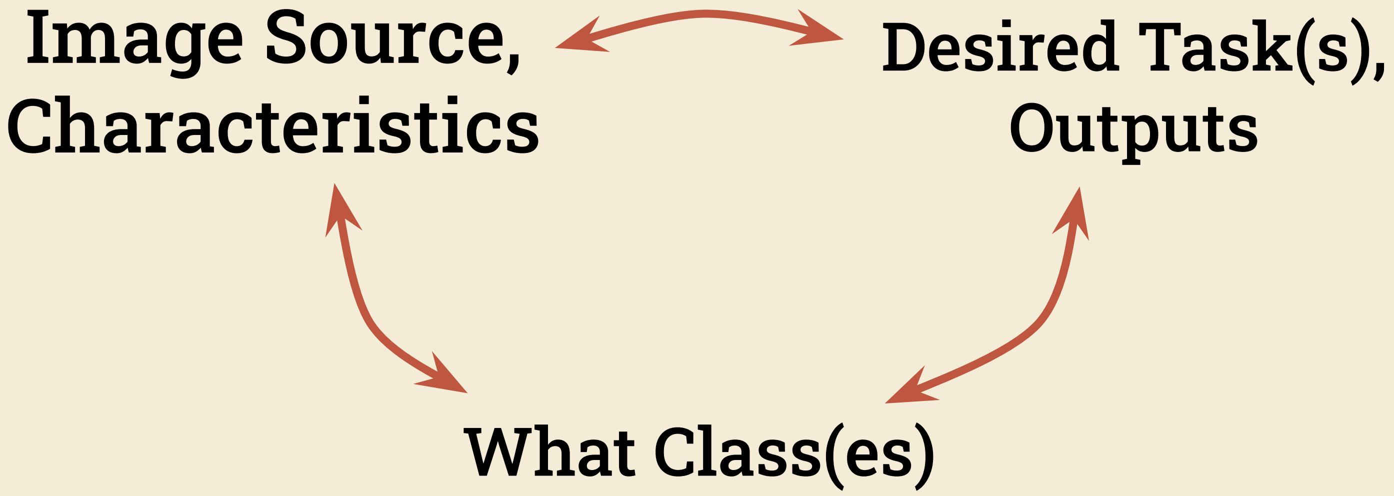
Assembling Images for Model Training



Include NULLS images that the model *may see*, but should ignore

NOT swamp milkweed...

Model Selection



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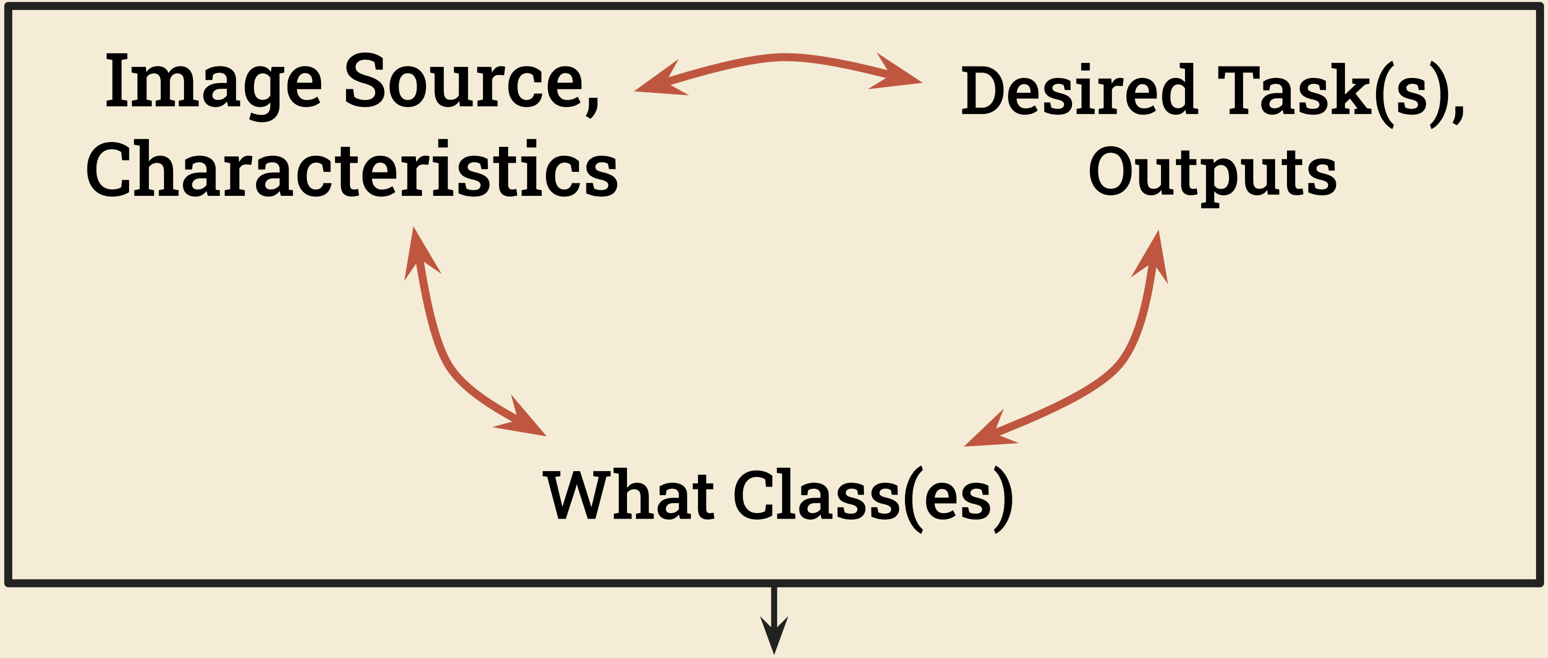
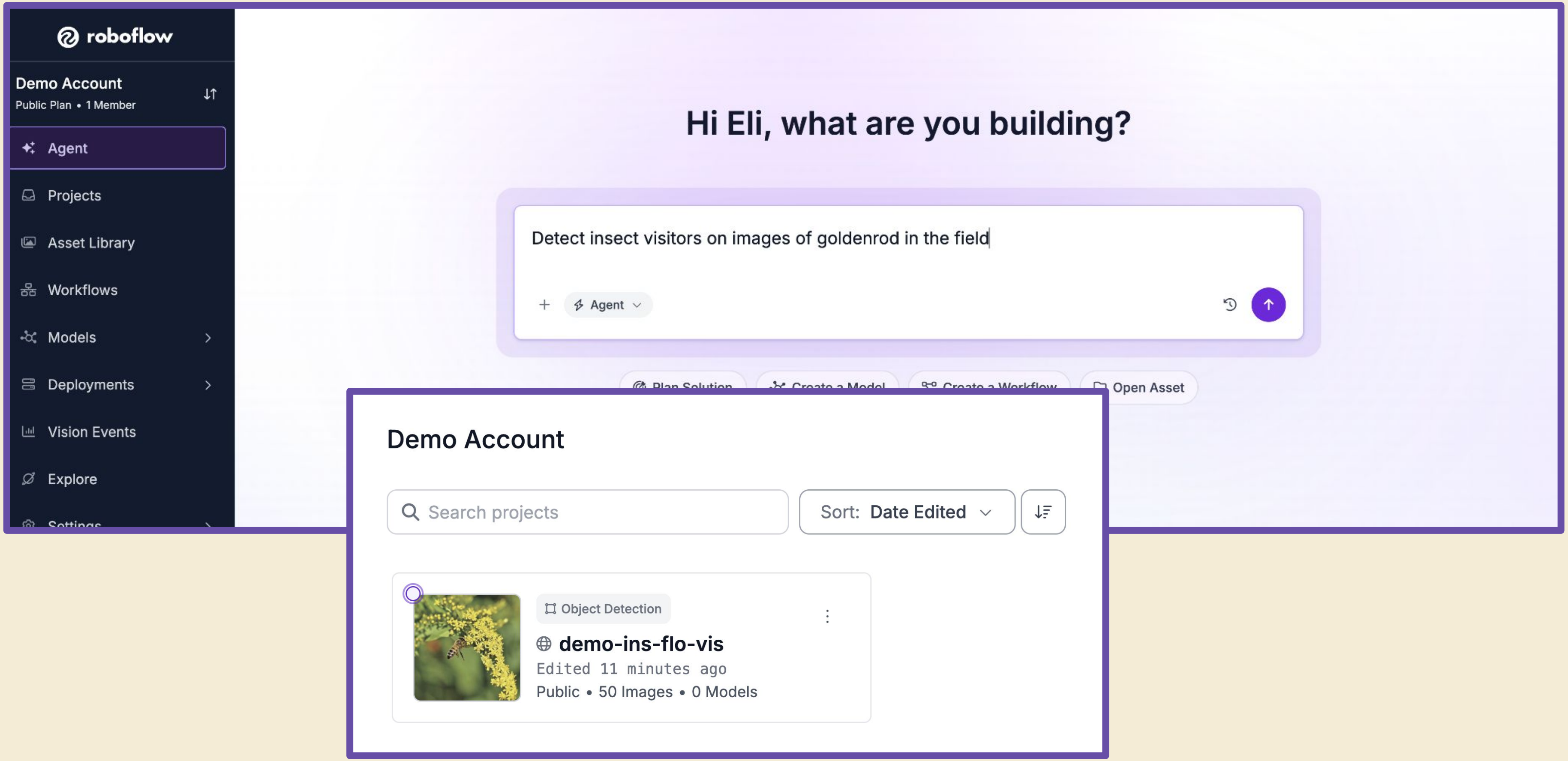


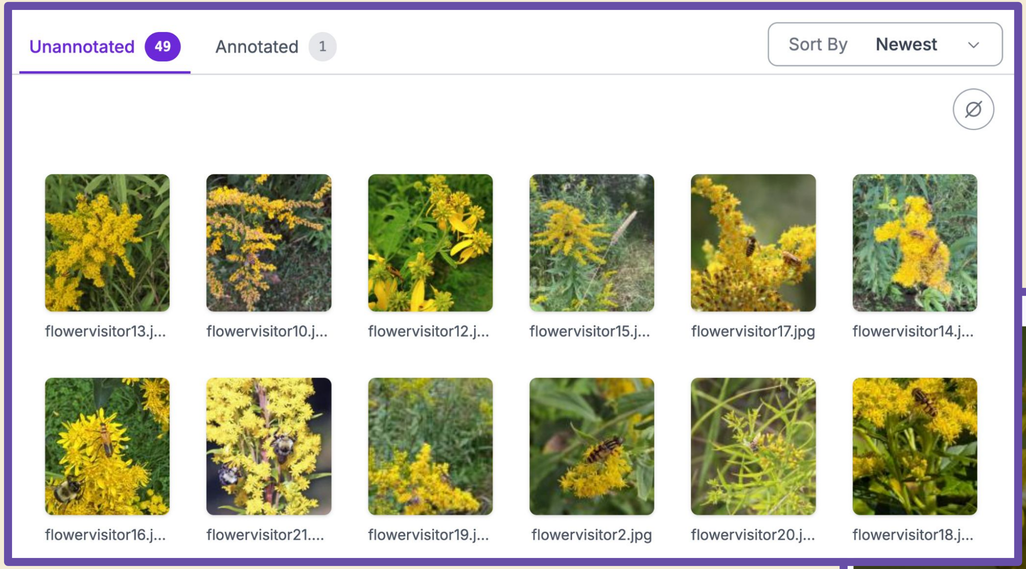
What Type of Model(s) You Need

this may change as you experiment...

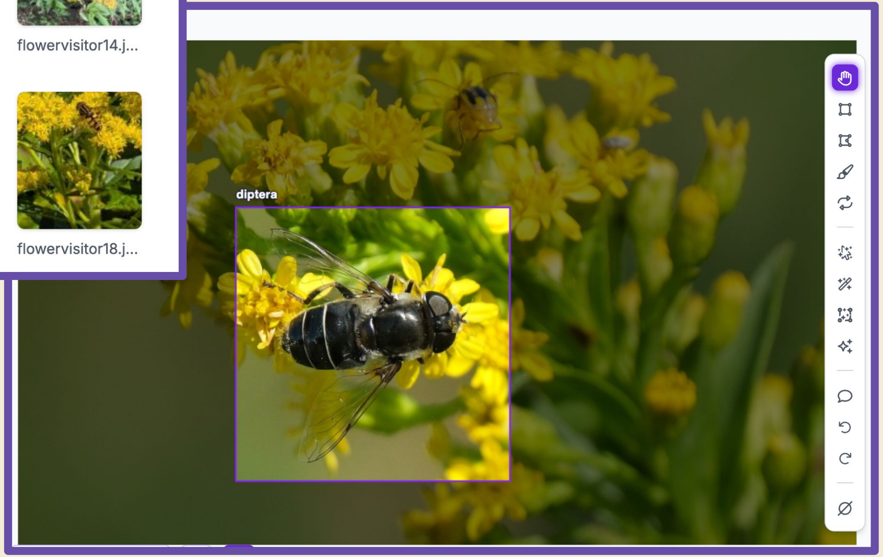
Annotating Images Using Roboflow

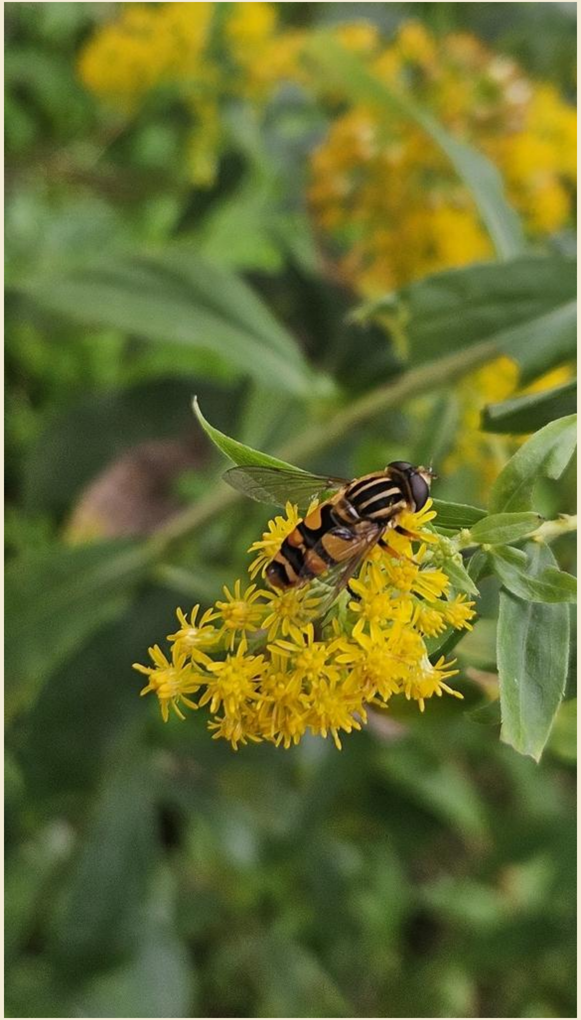


Nice Interface



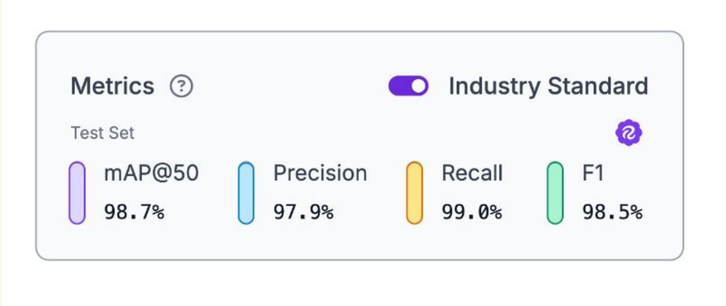
Lots of fun tools

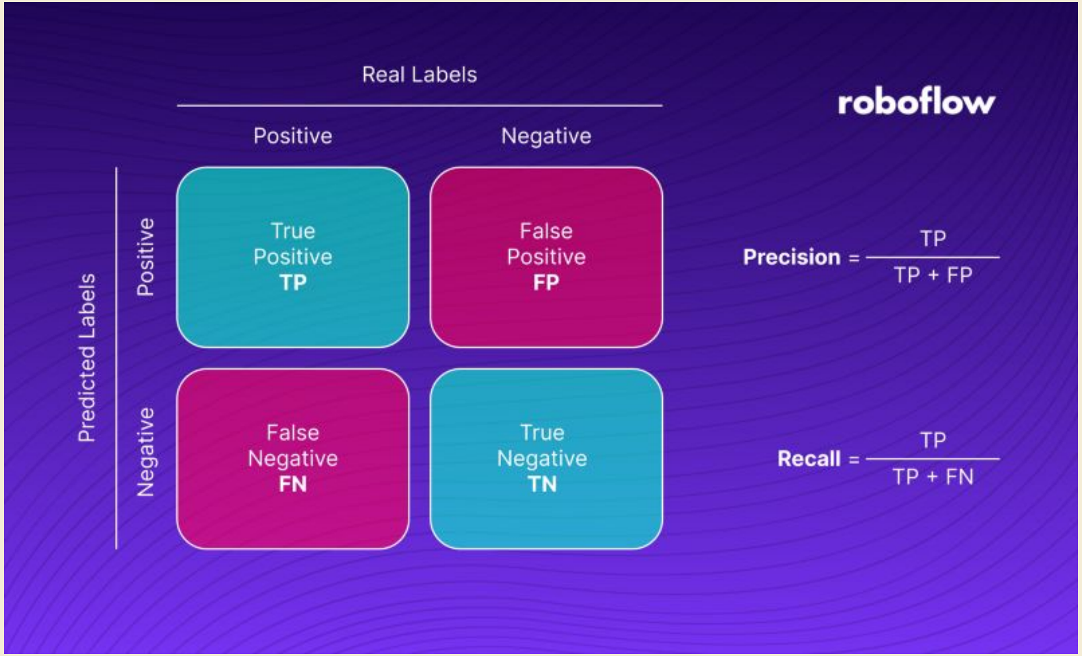


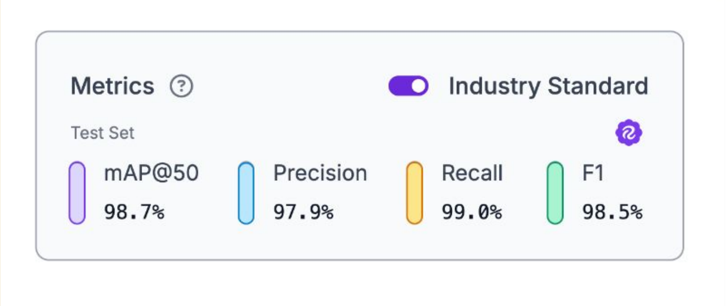


Let's train a model to help us measure:

Insect Abundance on Goldenrod


Metrics
Industry Standard
Test Set
mAP@50
98.7%
Precision
97.9%
Recall
99.0%
F1
98.5%


Real Labels
Positive
Negative
Predicted Labels
Positive
Negative
True Positive TP
False Positive FP
False Negative FN
True Negative TN
roboflow
Precision = TP / TP + FP
Recall = TP / TP + FN



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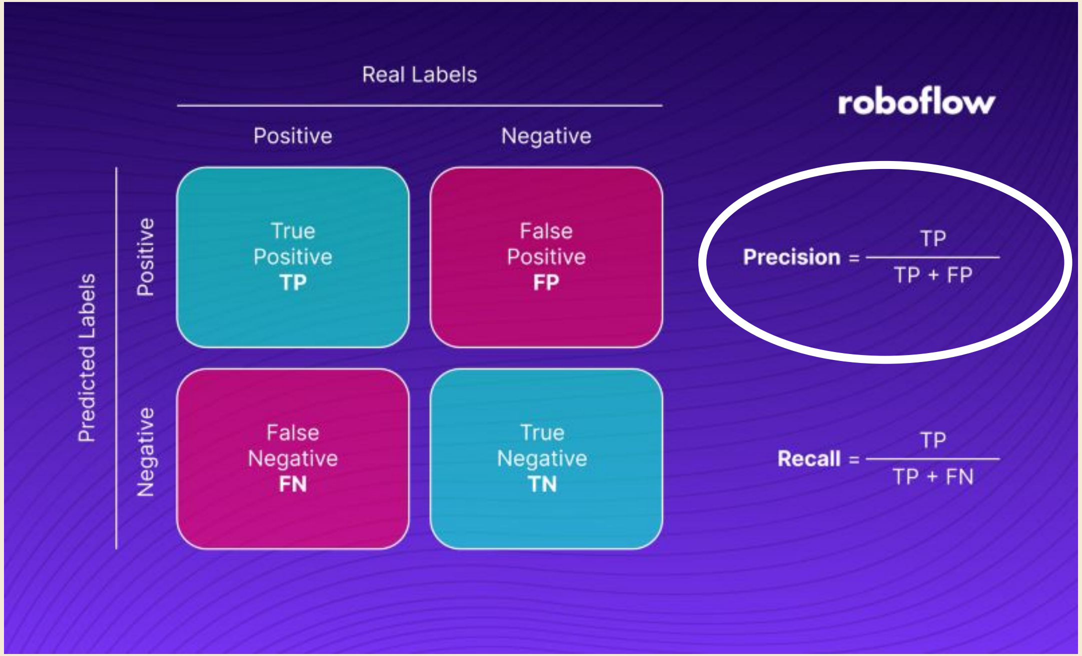

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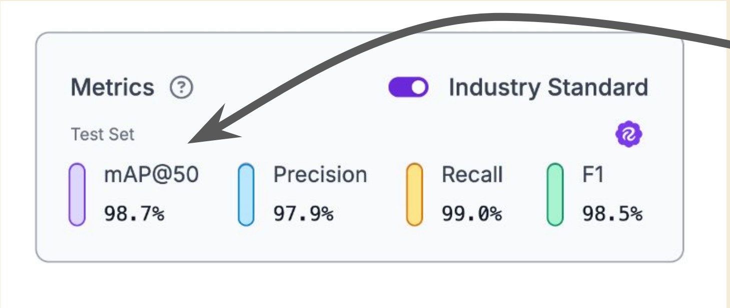
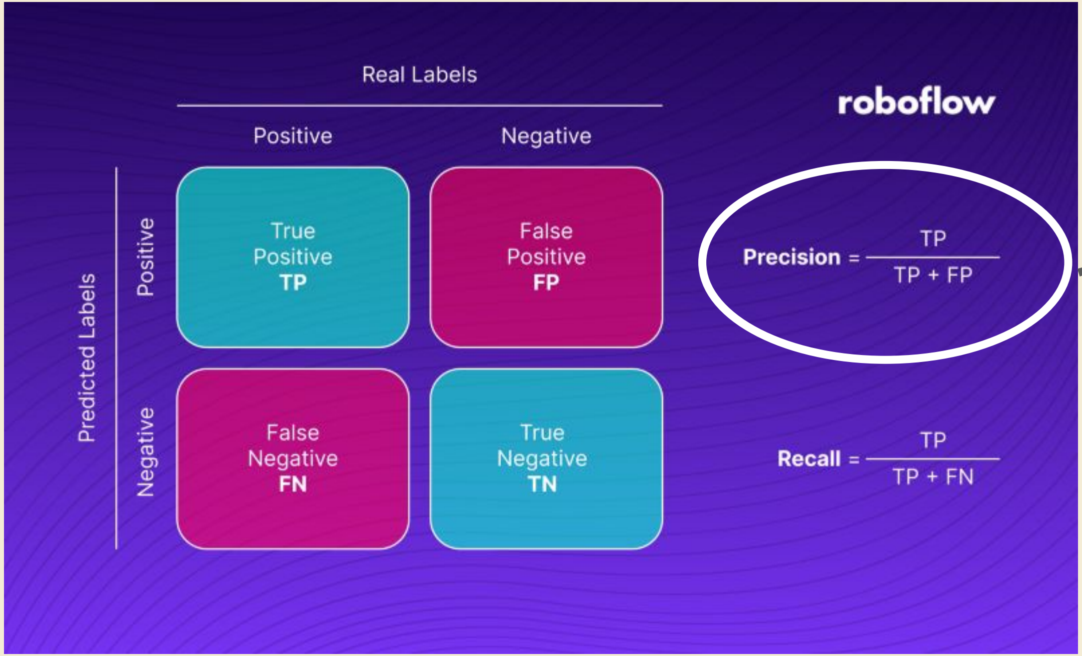
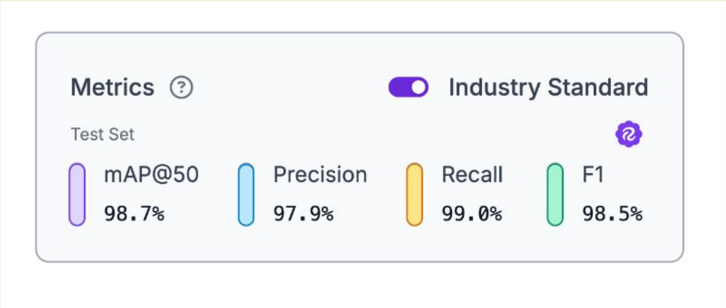




averaged across all classes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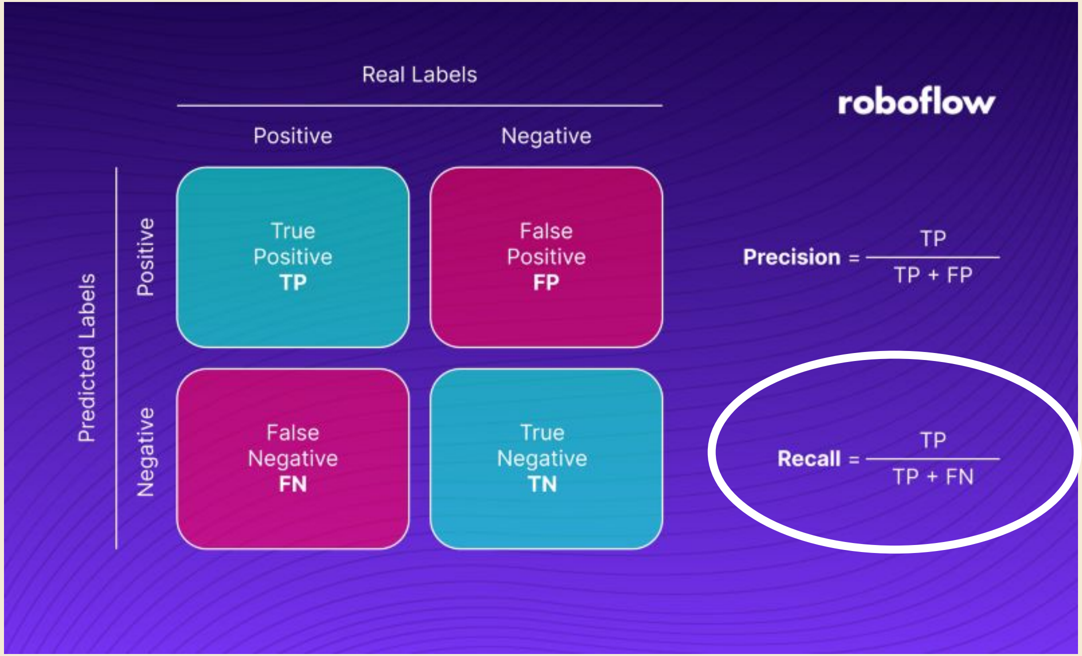
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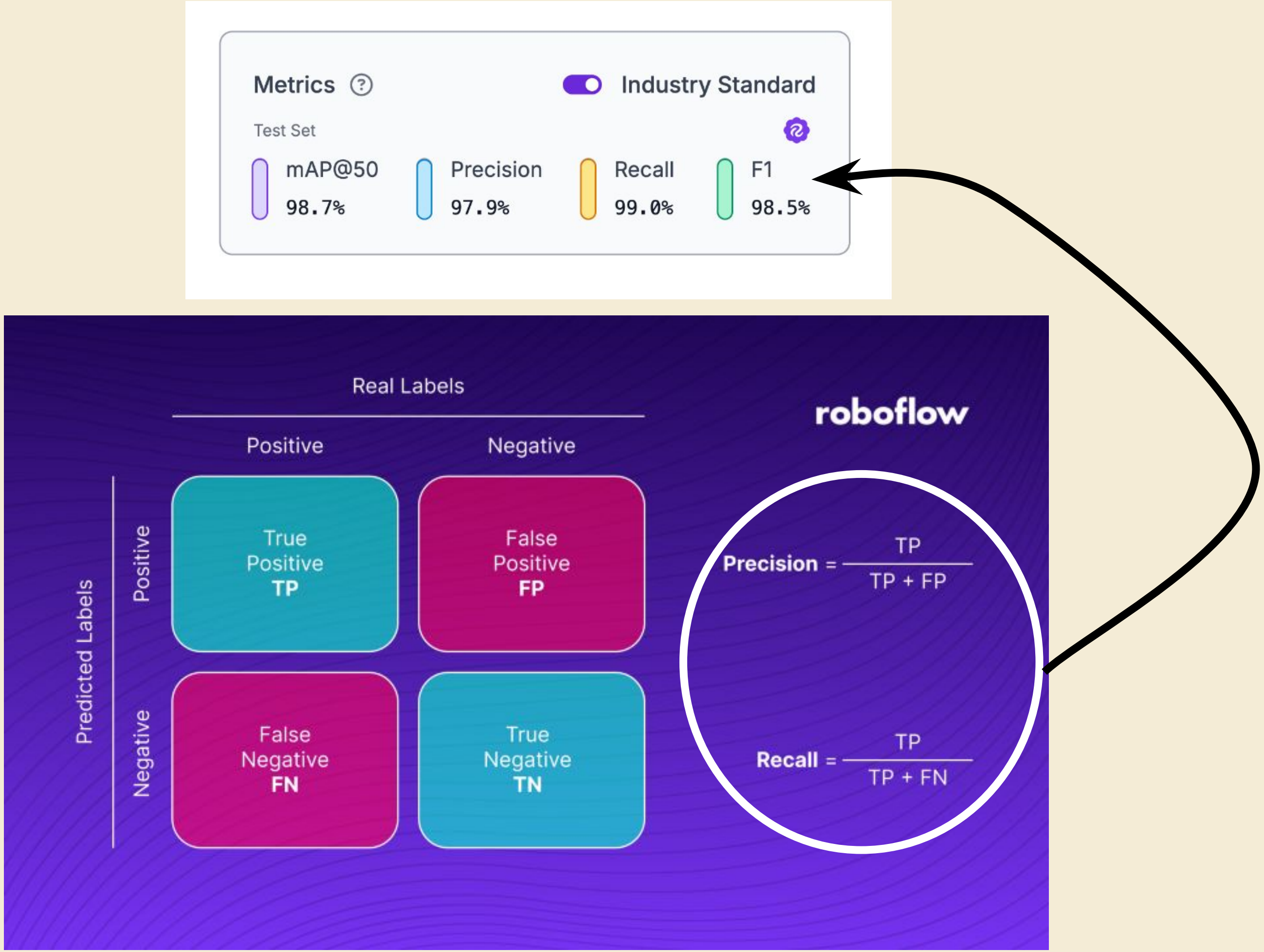
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